
Solver-Pinned Preference Data Teaches Small LLMs Strategic Play the Frontier Cannot Demonstrate

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Against a tit-for-tat opponent in repeated matrix games, Haiku 4.5, Gemma 3–27B,
2 and Llama 3.3–70B all average 1.50 cumulative reward per trajectory — exactly
3 the score of an untrained Qwen 2.5–3B. Frontier models reason fluently about
4 reciprocity and then fail to play it, which rules out the standard recipe of distilling
5 a larger teacher’s actions into a smaller student. We propose $A+\beta$, a preference-
6 data recipe built on the observation that the action and the reasoning in a game
7 trajectoretic setting have different best teachers: an analytical solver supplies each
8 round’s action label as the ex-post best response to the realised opponent move,
9 a frontier model paraphrases supporting reasoning under a prompt that forbids
10 leaking the unobserved current-round action, and an exact counterfactual rollout
11 against the recorded deterministic opponent discards flips whose myopic gain
12 would not survive the remaining horizon. On a twelve-environment suite spanning
13 eight in-domain and four structurally distinct held-out games, $A+\beta$ -trained Qwen
14 2.5–3B raises held-out reward from -1.45 to $+1.91$, 93% of Haiku 4.5’s $+2.05$
15 — a gain dominated by the auction, where solver-labelled pairs teach play that
16 frontier-derived data does not — and lifts tit-for-tat reward from 1.50 to 14.06.
17 Two specialist adapters trained on complementary data mixes recover each other’s
18 strengths under simple LoRA weight averaging, matching a reweighted retrain to
19 within 0.16 reward without any additional training. Across our experiments, the
20 provenance of the action label, not the scale of the teacher, determines whether
21 strategic play is learned.

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22 1 Introduction

23 Large language models underperform game-theoretic solution concepts in non-cooperative games
24 even when given room to reason in chain of thought beh [2025], gtb [2024]. The obvious training-time
25 remedy is preference optimization on game-play trajectories: collect rollouts, prefer high-reward
26 traces over low-reward ones, and let DPO Rafailov et al. [2023] push the policy toward better play.
27 Both halves of that recipe presume a teacher worth imitating — the trajectories come from a strong
28 model, and the preference signal comes from realised reward. This paper reports evidence that the
29 first presumption fails in a specific, measurable way, and develops the recipe that follows once it is
30 dropped.

31 The failure is easy to state. Against a tit-for-tat opponent in our repeated matrix games, Haiku 4.5,
32 Gemma 3–27B, and Llama 3.3–70B each average 1.50 cumulative reward per trajectory — exactly
33 the score of an untrained Qwen 2.5–3B (Table 4). Every one of these models defects early, triggers
34 retaliation, and never recovers, while writing fluent reasoning about reciprocity as it does so. The
35 failure is not confined to play: frontier models choose actions that contradict the argmax of their
36 own stated expected-value estimates in 15–70% of final rounds depending on model and game

(Appendix 5). A teacher that cannot play the game and does not follow its own reasoning is the wrong source of action labels, at any scale.

Our recipe, $A+\beta$, keeps the frontier model but demotes it. The observation underlying the design is that the action and the reasoning in a game trajectory are separate supervision targets with different best available teachers. For the action, an analytical solver provides the ex-post best response to the opponent’s realised move — provably optimal, and computable in closed form for matrix games, first-price auctions, and multi-issue bargaining. For the reasoning, the frontier model writes a justifying chain of thought under a prompt that forbids referencing the current-round opponent move the student will never observe at inference time. Between the two sits an exact counterfactual filter: because training opponents are deterministic heuristics, a dynamic-programming rollout decides exactly whether a myopically optimal flip would have improved the episode, and discards the flips — defect against the cooperator now — that trigger retaliation later. Section 2 develops the three components.

Trained with $A+\beta$ on 1749 preference pairs, Qwen 2.5–3B raises held-out reward from -1.45 to $+1.91$, which is 93% of Haiku 4.5’s $+2.05$ on the four games outside the original benchmark families, and raises tit-for-tat reward from 1.50 to 14.06 where every frontier reference remains at or below 1.50 (§3.2–3.3). The held-out gain is dominated by the auction, for which solver-labelled training pairs exist; the result demonstrates that solver labels teach a game frontier-action labels do not, rather than zero-shot transfer to unseen games (§3.2). Along the way we hit, and resolve twice over, a specialist-forgetting trade-off: a mix rich in auction data generalises but weakens strategic play, a mix rich in horizon-aware data does the reverse, and either upsampling the diluted slice $4\times$ or simply averaging the two specialists’ LoRA adapters — with no retraining — recovers both capabilities to within 0.16 reward of each other (§3.4). We also report what falsified our own starting hypothesis: filtering DPO pairs by reasoning-action coupling, with frontier actions as labels, made both coupling and reward *worse*; the gains appear only when the action label itself is replaced (§3.6).

Contributions. We make four contributions.

1. **$A+\beta$ (§2):** a DPO data-construction recipe that decouples action labels (analytical solver, ex-post best response) from reasoning labels (frontier paraphrase under an audited no-leakage prompt), filtered by exact counterfactual rollout, with a horizon-aware dynamic-programming variant for repeated games.
2. **A twelve-environment evaluation (§3.1):** eight in-domain and four held-out two-player games with per-opponent-class decomposition, under which we evaluate four frontier models and eight Qwen 2.5–3B variants (seven trained, plus base); the frontier-vs-TFT failure is, to our knowledge, undocumented at this granularity.
3. **Empirical findings (§3.2–3.3):** solver-pinned training recovers 93% of the frontier’s held-out reward from a 3B base and lifts strategic-opponent play the frontier itself cannot demonstrate, concentrated exactly where the frontier gap is largest.
4. **Specialist-adapter averaging (§2.5, §3.4):** two $A+\beta$ adapters trained on complementary mixes combine by parameter averaging to match a reweighted retrain within sampling noise — evidence that the specialists occupy linearly composable LoRA subspaces.

Scope. $A+\beta$ -trained 3B does not surpass Haiku on aggregate in-domain reward (13.93 vs. 14.60) or coupling (0.69 vs. 0.85), and all results concern one base model, one solver family, and deterministic training opponents; §3.7 states the boundaries plainly. The claim this paper defends is narrower and, we believe, more useful: when the task is strategic play, the provenance of the action label matters more than the scale of the model that produced it.

2 Method

$A+\beta$ is a data-construction recipe for preference optimization on game-play trajectories. Its design follows from a single diagnosis: in a game-play trajectory, the action and the reasoning that accompanies it are separate supervision targets with different best available teachers. The provably correct teacher for the action is an analytical solver; the best available teacher for the reasoning is a frontier LLM, provided it is prevented from leaking information the student will not have at inference time; and neither teacher, on its own, knows whether a locally optimal action helps or hurts over the

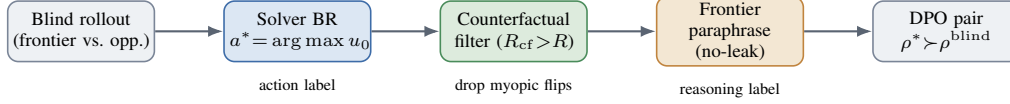


Figure 1: $A+\beta$ data construction. A blind rollout produces the rejected side. The analytical solver supplies the *action* label (provably-optimal best response to the realised opponent move); a counterfactual horizon rollout drops flips whose myopic optimality would not improve episode reward; the frontier model supplies the *reasoning* label under a no-leakage posterior-EV prompt. The chosen side pairs the solver action with the frontier reasoning.

88 remaining horizon, which a counterfactual rollout against the recorded opponent can decide exactly.
 89 $A+\beta$ assigns each of these three jobs to the component that performs it provably or measurably
 90 best. This section fixes notation (§2.1), then develops the three components — solver-pinned best
 91 response (§2.2), no-leakage frontier paraphrase (§2.3), and the counterfactual horizon filter (§2.4)
 92 — and closes with specialist-adaptor averaging (§2.5) and the assembled pipeline (§2.6). Figure 1
 93 sketches the data flow.

94 2.1 Notation and setup

95 We work with two-player games $G = (\mathcal{A}_0, \mathcal{A}_1, u_0, u_1)$ in two modes. One-shot games (auctions,
 96 multi-issue bargaining, single-stage matrix games) terminate after one joint action. Repeated games
 97 of horizon T (iterated prisoner’s dilemma variants, repeated stag hunt) play the same stage game T
 98 times, with each player observing the joint history before each round. We take the perspective of
 99 player 0; player 1 is the opponent.

100 A trajectory $\tau = (x, \rho, \mathbf{a})$ contains a game context x , a chain-of-thought reasoning string ρ_t at
 101 each round t , and an action sequence $\mathbf{a} = (a_t^0)_{t < T}$. The realised cumulative reward is $R(\tau) =$
 102 $\sum_t u_0(a_t^0, a_t^1)$, where a_t^1 is the opponent’s action at round t , produced by a deterministic heuristic
 103 policy π_{opp} during training-data generation. DPO Rafailov et al. [2023] consumes preference pairs
 104 (x, y_w, y_l) in which y_w is preferred to y_l given context x .

105 The natural preference signal for game play is realised cumulative reward, but on a single rollout
 106 that scalar entangles three distinct sources of credit: which rounds were reasoned well, which were
 107 reasoned poorly, and how the opponent’s policy amplified or forgave each choice. Whole-trajectory
 108 reward pairing therefore rewards and punishes rounds indiscriminately. $A+\beta$ instead constructs
 109 preferences at round granularity, using the three components that follow.

110 2.2 Solver-pinned best response

111 The first component replaces the acting model’s action with a label whose correctness is a property
 112 of the game rather than of any model. For each round t of a recorded trajectory, we substitute
 113 the recorded action with the best response to the opponent’s actually-realised move a_t^1 , computed
 114 analytically:

$$\text{BR}_0(a_t^1) = \arg \max_{a \in \mathcal{A}_0} u_0(a, a_t^1). \quad (1)$$

115 Conditioning on the realised a_t^1 makes Eq. 1 an ex-post label: it asks what would have maximised
 116 this round’s payoff given what the opponent in fact did, which is computable exactly after the fact
 117 even though it is not observable at decision time. §2.3 addresses the resulting information asymmetry
 118 between label construction and inference.

119 We implement Eq. 1 for three game families. For two-player matrix games the solver enumerates the
 120 $|\mathcal{A}_0|$ own-action payoffs against the recorded opponent column and returns the argmax, at $O(|\mathcal{A}_0|)$
 121 cost. For first-price sealed-bid auctions with private values $v_0, v_1 \sim \text{Unif}[0, v_{\max}]$, the ex-post best
 122 response to the opponent’s bid b_1 is the smallest legal bid strictly above b_1 when $v_0 > b_1 + \epsilon$, and
 123 zero (a deliberate loss) otherwise. For multi-issue bargaining with capacity c across three issues and
 124 own weights w_0 , the ex-post best response to the opponent’s proposal $\mathbf{p}_1$ is the take-the-rest proposal
 125 $p_0[i] = c - p_1[i]$, which maximises u_0 subject to joint feasibility.

126 **Why not distil the frontier’s actions.** The obvious alternative — take a strong frontier model’s play
 127 as the label — fails for a measurable reason. Frontier LLMs produce sophisticated reasoning traces,
 128 complete with posterior beliefs and expected-value arithmetic, and then choose actions that deviate
 129 from the argmax of their own stated expected values — in 15% of final rounds for the best case
 130 (Haiku 4.5 in-domain) and up to 70% in the worst (Llama 3.3–70B held-out), with per-game rates
 131 tabulated in Appendix 5. Distilling frontier actions therefore bakes this reasoning-action decoupling
 132 into the student. The solver label costs no more than the legal-action enumeration and is exact by
 133 construction; the empirical consequence of the swap is the subject of §3.

134 2.3 No-leakage frontier paraphrase

135 The solver outputs an action but no reasoning, and a DPO pair whose chosen side contains only an
 136 action string would supervise none of the chain-of-thought tokens that dominate the student’s output
 137 distribution. We therefore use a frontier model (Bedrock Haiku 4.5) to write the reasoning ρ_t^* that
 138 justifies the solver action a^* — and here the ex-post nature of the label becomes a liability. The label
 139 conditions on the opponent’s current-round action; the student at inference time observes only the
 140 history through round $t - 1$. Reasoning that says “the opponent defected this round, so I defect”
 141 would train the student to expect an observation that never exists at decision time.

142 We resolve the asymmetry by prompt construction rather than by information removal. The para-
 143 phraser is told the realised opponent action and the solver’s best response out of band, but is
 144 constrained to write reasoning expressible from history alone — belief-state reasoning over the
 145 opponent’s type whose posterior happens to support a^* :

146 SYSTEM. You are documenting strategic decisions . . . The other player’s current-round
 147 move is NOT observable at decision time. We will tell you (out-of-band, for your reference)
 148 the opponent’s actual move that round and the solver-computed best response, but your
 149 written reasoning must NEVER reference the current-round opponent move as observed.
 150 Treat it as belief-state reasoning under uncertainty. . . . Hard rules: (1) NEVER write
 151 phrases like “opponent played C this round”; (2) The [EV] arithmetic must average over
 152 your posterior types — not condition on the unrevealed action; (3) Your [DECISION] must
 153 match the solver-pinned best response we provide; build the posterior to justify that action.

154 We audit the constraint rather than assume it holds. Across 688 oracle rounds in the final data batches,
 155 no round-1 trace references the current opponent action, and the 1% of rounds matched by a heuristic
 156 leakage regex are all, on manual inspection, legitimate discussion of prior rounds (Appendix 5).

157 2.4 Counterfactual horizon filter

158 The solver label of §2.2 is myopic: against any cooperator, it says to defect now, whatever the future
 159 holds. In repeated games against reciprocity opponents such as tit-for-tat or grim trigger, a round-1
 160 defection triggers retaliation that costs far more over the horizon than the single-round gain, so pairing
 161 every solver flip as a chosen example would teach exactly the defection bias that makes frontier
 162 models fail against these opponents (§3.3).

163 The filter admits a flip only if it would have improved the episode. Given a recorded blind trajectory
 164 τ_{blind} and a candidate flip to a^* at round t , we recompute the episode reward with the flip inserted
 165 and the opponent’s recorded policy replayed against the perturbed history:

$$R_{\text{cf}}(t, a^*) = \sum_{s < t} u_0(a_s^0, a_s^1) + u_0(a^*, a_t^1) + \sum_{s > t} u_0(a_s^0, \pi_{\text{opp}}(\mathbf{h}_s^*)), \quad (2)$$

166 where $\mathbf{h}_s^*$ denotes the counterfactual history including the flip. Because our training opponents are
 167 deterministic heuristics whose policies are reconstructible from their names, Eq. 2 is exact rather than
 168 estimated. A DPO pair $(\rho_t^* \succ \rho_t^{\text{blind}})$ is emitted only when $R_{\text{cf}}(t, a^*) > R(\tau_{\text{blind}})$.

169 **Horizon-aware variant.** Eq. 2 holds the student’s post-flip actions fixed at their recorded values,
 170 which understates the value of a flip that opens better continuations. The stronger counterfactual
 171 asks what the episode would be worth if play after the flip were optimal. Let $V(s, \mathbf{h}_s)$ denote the
 172 maximum attainable cumulative reward from round s under history $\mathbf{h}_s$; it satisfies the dynamic
 173 program

$$V(s, \mathbf{h}_s) = \max_{a \in \mathcal{A}_0} \left[u_0(a, \pi_{\text{opp}}(\mathbf{h}_s)) + V(s + 1, \mathbf{h}_s \oplus (a, \pi_{\text{opp}}(\mathbf{h}_s))) \right] \quad (3)$$

with $V(T, \cdot) = 0$, and the horizon-aware counterfactual reward becomes $R_{\text{cf}}^{\text{HA}}(t, a^*) = \sum_{s < t} u_0(a_s^0, a_s^1) + u_0(a^*, a_t^1) + V(t+1, \mathbf{h}_{t+1}^*)$. For two-action matrix games at horizon $T \leq 10$, exhaustive enumeration of the 2^{T-t} continuations is cheap enough that no pruning is needed. The filter thereby converts DP-optimality against a known opponent into a training-time preference signal, and it is the component that separates “defect on the cooperator now” from “cooperate now because retaliation is coming” (§3.3).

Relation to K -sample credit assignment. GRPO Shao et al. [2024] addresses the same credit-assignment problem — which round deserves the episode’s reward — by sampling K on-policy rollouts and normalising terminal rewards within the group. When the opponent’s policy is known and deterministic, a single DP pass yields the exact return of every continuation, information that K -sample on-policy estimates approach only stochastically and at K rollouts of cost per state. The trade is generality for exactness: GRPO applies against any opponent, while Eq. 3 requires the opponent to be reconstructible from its name, a condition our training regime satisfies by construction and deployment against unknown opponents does not.

2.5 Specialist-adapter averaging

Training $A+\beta$ on different data mixes produces specialists with complementary strengths and a shared geometry. In our experiments, the AUX mix (auction- and negotiation-heavy) generalises best out of domain while the ALL mix (horizon-aware-PD heavy) plays strategic in-domain opponents best (§3.4); both adapters are trained with identical LoRA configuration ($r=16$, $\alpha=32$, identical target modules) on the same base model, so their weights are coordinates in the same adapter space. We combine them by direct parameter averaging:

$$W_{\text{merge}} = \alpha W_{\text{AUX}} + (1 - \alpha) W_{\text{ALL}}, \quad (4)$$

applied to each of the 504 LoRA tensors with $\alpha = 0.5$, with no subsequent fine-tuning. The merged adapter loads exactly as either parent. This is task arithmetic Ilharco et al. [2023], Yadav et al. [2023] applied to game-theoretic specialists, and its empirical claim is deliberately narrow: for these two adapters, §3.4 shows the average matches a $4\times$ reweighted retrain within sampling noise, indicating that the two specialists occupy linearly composable subspaces rather than conflicting ones.

2.6 Assembled pipeline

Algorithm 1 assembles the three components over one episode: roll out the blind trajectory, pin each round’s action to the solver, keep only flips the counterfactual certifies, and paraphrase reasoning for the survivors.

Algorithm 1 $A+\beta$ data construction (per episode)

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1: Input: game  $G$ , opponent policy  $\pi_{\text{opp}}$ , frontier paraphraser  $\text{LLM}_{\text{front}}$ , horizon-aware flag  $h$ 
2: Roll out blind trajectory  $\tau_{\text{blind}} = (x, \rho^{\text{blind}}, \mathbf{a}^{\text{blind}})$  with  $\text{LLM}_{\text{front}}$  as player 0
3: for round  $t = 0, \dots, T-1$  do
4:    $a^* \leftarrow \text{BR}_0(a_t^1)$  ▷ Eq. 1
5:   if  $a^* \neq a_t^{0, \text{blind}}$  then
6:      $R_{\text{cf}} \leftarrow$  horizon-aware Eq. 3 if  $h$  else Eq. 2
7:     if  $R_{\text{cf}} > R(\tau_{\text{blind}})$  then
8:        $\rho_t^* \leftarrow \text{LLM}_{\text{front}}(\text{no-leak prompt}, a^*)$ 
9:       Emit pair  $(x_t, \rho_t^* \oplus a^* \succ \rho_t^{\text{blind}} \oplus a_t^{0, \text{blind}})$ 
10:    end if
11:  end if
12: end for

```

The student (Qwen 2.5–3B-Instruct) is trained on the emitted pairs with LoRA ($r=16$, $\alpha=32$, dropout 0.05), sigmoid DPO loss with $\beta = 0.1$, and environment-token loss masking, so that the student is not penalised for reproducing environment-emitted observation strings. Checkpoints are selected by held-out DPO eval loss. Full hyperparameters appear in Appendix 5.

3 Experiments

Our evaluation is organised around one question: does replacing frontier-model action labels with solver best responses give a small model strategic competence that neither its own scale nor the frontier teacher possesses? We first describe the evaluation suite (§3.1), then present the headline held-out result (§3.2), locate where the gain concentrates (§3.3), analyse the specialist-forgetting trade-off and its two equivalent resolutions (§3.4), report reasoning-action coupling as a complementary diagnostic (§3.5), document the falsified precursor hypothesis (§3.6), and close with scope and limitations (§3.7).

3.1 Setup

Environments. We evaluate on twelve two-player environments, split into eight in-domain (ID) games and four held-out (HO) games. The split was fixed at benchmark-design time: the ID set contains the repeated-game families that dominate training, and the HO set contains structurally distinct game types. One HO game, the auction, later received solver-labelled training pairs in the broader $A+\beta$ mixes (§3.4); the remaining three HO games, and three of the ID games (Battle of the Sexes, Matching Pennies, tic-tac-toe), appear in no training mix. The ID games comprise three iterated prisoner’s dilemma payoff variants (pd-classic $T=5/R=3/P=1/S=0$, pd-tight $T=4/R=3/P=2/S=1$, pd-high-temptation $T=8/R=3/P=1/S=0$, each with horizon 10), repeated stag hunt, multi-issue bargaining, Battle of the Sexes, Matching Pennies, and tic-tac-toe. The HO games are a first-price sealed-bid auction with uniform private values ($v_{\max} = 100$), divide-the-dollar, K -level p -beauty, and a one-shot prisoner’s dilemma stage game. Each (model, environment) pair is evaluated over $n = 12$ episodes against a per-environment opponent pool: nine deterministic heuristics for the repeated matrix games, and four bidder and four bargainer heuristics for the auction and bargaining families respectively (Appendix 5).

Models. All trained variants fine-tune Qwen 2.5–3B-Instruct. Two variants realise the falsified precursor recipe of §3.6 (coupling-filtered DPO on frontier actions, filter on and off); five realise $A+\beta$ on increasingly broad data mixes: CORE (503 matrix-game pairs), AUX (613 pairs, auction- and negotiation-heavy), ALL (1338 pairs, dominated by horizon-aware PD data), RW (the ALL mix with auction and negotiation pairs upsampled 4 $\times$), and MERGE (the parameter average of the AUX and ALL adapters, Eq. 4, with no additional training). As frontier references we evaluate Bedrock Haiku 4.5 — the same model that produces $A+\beta$ ’s reasoning paraphrases — together with Gemma 3–27B, Llama 3.3–70B, and Llama 3.1–8B.

Metrics. The primary metric is cumulative reward (CR) per trajectory, averaged over episodes and opponents, reported separately for ID and HO games. As a secondary diagnostic we measure reasoning-action coupling: the fraction of parsable rounds in which the chosen action equals the argmax of the expected values the model itself states in its chain of thought. We report final-round coupling (fc) and all-round coupling (ac); the parser is described in Appendix 5.

Training. All variants use LoRA ($r=16$, $\alpha=32$, dropout 0.05) with sigmoid DPO loss ($\beta=0.1$), environment-token loss masking, learning rate 5×10^{-5} with cosine decay, 10 epochs, and best-checkpoint selection by held-out DPO eval loss. Training runs on a single L40S GPU within 13 GB of memory, and wall-clock time scales linearly with pair count: 60 minutes for the 503-pair CORE mix, 219 minutes for the 1338-pair ALL mix. Full hyperparameters are given in Appendix 5.

3.2 Held-out reward: solver labels recover what scale does not

Table 1 reports aggregate ID and HO cumulative reward for the eight Qwen 2.5–3B variants and the four frontier references; Figure 2 plots the same models in (ID, HO) reward space. The headline result is the held-out column. The base model loses reward on held-out games (CR -1.45), and no variant of the precursor recipe brings it above zero. $A+\beta$ -RW reaches $+1.91$ and $A+\beta$ -MERGE $+1.75$, against Haiku 4.5’s $+2.05$: the trained 3B model attains 93% of the held-out reward of the frontier model whose own actions it was *not* trained to imitate ($1.91/2.05$), while giving up almost none of its in-domain reward (13.93 vs. base 14.04). Two caveats scope this number. First, the HO gain is dominated by the auction, which the RW mix trains on with solver-labelled pairs (§3.1); the result shows that solver labels teach a game frontier-derived data does not, rather than zero-shot transfer — the three strictly unseen HO games move little in either direction (Table 2). Second, because the

Table 1: Headline results. Qwen 2.5–3B trained variants vs. frontier references on 12-env eval (8 ID + 4 HO, $n = 12$ episodes per env). CR = mean cumulative reward. Reasoning-action coupling is reported separately in §3.5 (Table 5). Best 3B-variant cell per metric in **bold**.

Model	CR(ID)	CR(HO)
Qwen 2.5–3B base	14.04	−1.45
Qwen 2.5–3B + DPO (filter-on)	13.77	−0.95
Qwen 2.5–3B + DPO (filter-off)	13.51	−0.23
Qwen 2.5–3B + A+ β -CORE	13.79	+0.44
Qwen 2.5–3B + A+ β -AUX	13.86	+0.68
Qwen 2.5–3B + A+ β -ALL	14.34	−0.27
Qwen 2.5–3B + A+ β -RW	13.93	+1.91
Qwen 2.5–3B + A+ β -MERGE	13.92	+1.75
Bedrock Haiku 4.5	14.60	+2.05
Gemma 3–27B	13.36	+0.28
Llama 3.3–70B	12.29	+1.52
Llama 3.1–8B (Bedrock)	13.60	−0.06

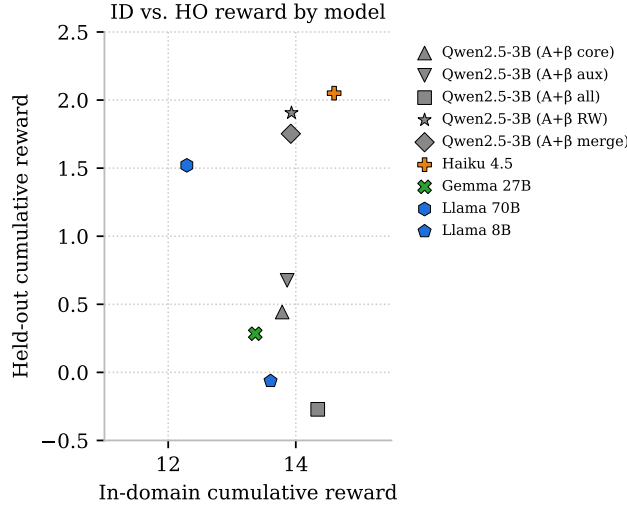


Figure 2: In-domain vs. held-out cumulative reward by model. Qwen 2.5–3B A+ β variants are shown in grey (distinguished by marker); frontier reference models are coloured (Haiku orange, Gemma green, Llama blue). A+ β -RW (star) and A+ β -MERGE (diamond) sit in the upper region near Haiku 4.5, recovering held-out reward without sacrificing in-domain reward.

259 solver, not Haiku, supplies every action label, the recovery cannot be explained as distillation of
 260 frontier play; the frontier contributes only paraphrased reasoning (§2.3).

261 3.3 The gain concentrates where the frontier gap is largest

262 Two decompositions locate the held-out gain. Per game (Table 2), the auction dominates: base CR
 263 −8.67 rises to +4.97 for RW, against Haiku’s +5.82, and no other single game moves comparably.
 264 Per opponent class (Table 3), we group the nine opponents into stationary (always-cooperate, always-
 265 defect), stochastic (random, probabilistic), and strategic (tit-for-tat, grim trigger, Pavlov, greedy
 266 tic-tac-toe) axes. On stationary opponents the base model already plays well (ID CR 16.60) and
 267 training changes little. On stochastic opponents the held-out picture is the informative one: base
 268 loses heavily (−3.33) where RW reaches +2.23, approaching Haiku’s +2.67. On strategic opponents
 269 the ID lift concentrates: base 13.26 rises to 14.62 (RW) and 15.54 (MERGE), against Haiku’s 21.13.
 270 The two zero-sum ID games (Matching Pennies, tic-tac-toe) score zero for every model and do not
 271 separate the variants under this metric.

Table 2: Per-game cumulative reward. Trained 3B variants and Haiku 4.5 reference. Best 3B cell per row **bold**.

Game	base	core	aux	all	rw	merge	Haiku
<i>In-domain (ID)</i>							
pd-classic	21.96	22.08	23.08	23.88	21.67	23.50	25.68
pd-tight	25.92	27.08	26.21	26.21	25.42	25.92	27.25
pd-high-temptation	29.33	28.50	29.17	30.42	28.42	29.17	27.92
stag-hunt	28.62	28.71	28.38	30.54	30.42	27.88	27.79
negotiation	5.75	3.17	3.46	3.08	4.67	4.04	7.54
bos	0.75	0.75	0.62	0.62	0.88	0.88	0.62
matching-pennies	0.00	0.00	0.00	0.00	0.00	0.00	0.00
tic-tac-toe	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Held-out (HO)</i>							
auction	-8.67	-0.71	-0.10	-3.96	4.97	4.18	5.82
divide-dollar	0.27	0.11	0.22	0.34	0.32	0.21	0.25
p-beauty	0.50	0.50	0.50	0.50	0.50	0.50	0.50
ipd-stage	2.08	1.88	2.08	2.04	1.83	2.12	1.62

Table 3: Per-opponent-axis CR (all 12 envs combined, $n = 12$ episodes per env). Opponents grouped by behavioural axis: stationary (always-C, always-D), stochastic (random, probabilistic), strategic (TFT, grim, paylov, ttg-greedy). **Strategic-opponent CR is where the lift concentrates**: base 13.26, A+ β -RW 14.62, Haiku 21.13.

Set	Axis	base	core	aux	all	rw	merge	Haiku
ID	Stationary	16.60	16.19	15.84	16.35	16.40	15.92	18.79
ID	Stochastic	7.22	5.53	6.66	6.21	6.37	6.34	6.83
ID	Strategic	13.26	15.12	15.25	16.35	14.62	15.54	21.13
HO	Stationary	1.41	1.29	1.48	1.48	1.37	1.49	1.10
HO	Stochastic	-3.33	-0.07	0.20	-1.32	2.23	1.91	2.67

The single most instructive opponent is tit-for-tat (Figure 3, right). Against TFT the base model scores 1.50, and so do Haiku 4.5, Gemma 3–27B, and Llama 3.3–70B — every frontier reference defects early, triggers retaliation, and never recovers (Table 4). A+ β -RW scores 14.06. We attribute the lift to the counterfactual filter, the one component designed to reject flips whose myopic gain destroys future reward (§2.4); consistent with this, every A+ β mix — including CORE, which predates the horizon-aware variant — lifts TFT-opponent reward to at least 12.88, while both precursor variants trained on frontier actions leave it unchanged. A component ablation that removes the filter while keeping solver labels would isolate the mechanism directly and is left to future work. Parameter count does not substitute for this signal, because the failure is shared by models from 3B to 70B and by the frontier teacher itself; scale buys fluent reasoning about reciprocity, not reciprocal play.

3.4 Reweighting and adapter averaging resolve the same trade-off

The A+ β mixes reveal a specialist-forgetting trade-off before resolving it. AUX devotes 18% of its 613 pairs to auction and negotiation data and wins held-out reward (+0.68), but is the weakest strategic ID player (TFT-opponent CR 12.88 against ALL’s 15.31). ALL dilutes that share to 10% (137 of 1338 pairs), wins ID strategic play, and collapses on the auction (CR -0.10 for AUX falls to -3.96). The dilution of one subdistribution, not any change in recipe, causes the forgetting.

We resolve the trade-off in two independent ways. The data-side intervention (RW) upsamples the 137 auction and negotiation pairs $4\times$ into a 1749-pair mix and retrains. The parameter-side intervention (MERGE) averages the AUX and ALL adapters (Eq. 4, $\alpha = 0.5$) with no retraining. Table 1 shows the two land within 0.16 held-out CR of each other (+1.91 vs. +1.75) and within 0.01 ID CR, differences we cannot distinguish from noise at $n = 12$ episodes per environment (§3.7). A practitioner holding two specialist adapters can therefore obtain the reweighted-retrain result by averaging weights, at the cost of a tensor sum rather than 219 GPU-minutes of training. The equivalence also suggests the two

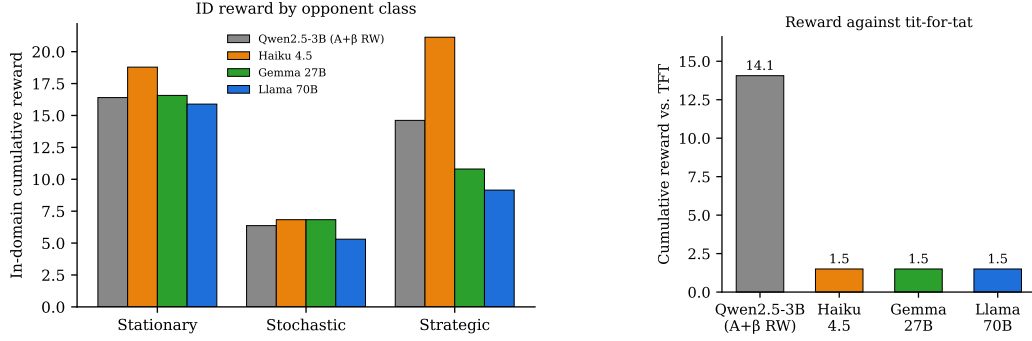


Figure 3: Left: in-domain reward by opponent class for the trained Qwen 2.5–3B A+β-RW model against three frontier references (Haiku 4.5, Gemma 3–27B, Llama 3.3–70B). On stationary and stochastic opponents all four models are bunched; on *strategic* opponents A+β-RW (14.62) clears the larger frontier models (Gemma 10.80, Llama-70B 9.15) and trails only Haiku (21.13). Right: reward against tit-for-tat specifically — A+β-RW reaches 14.06, while all three frontier references score only 1.50.

Table 4: Frontier reference rows on the same 12-env eval. Useful as a benchmark in itself. CR = mean cumulative reward; TFT-CR = cumulative reward against the tit-for-tat opponent. Coupling for these models is reported in Table 5.

Model	CR(ID)	CR(HO)	TFT-CR
Bedrock Haiku 4.5	14.60	+2.05	1.50
Gemma 3–27B	13.36	+0.28	1.50
Llama 3.3–70B	12.29	+1.52	1.50
Llama 3.1–8B (Bedrock)	13.60	−0.06	0.38
Qwen 2.5–3B base	14.04	−1.45	1.50

specialists occupy linearly composable LoRA subspaces rather than conflicting ones; testing whether this survives more than two specialists is future work.

3.5 Reasoning-action coupling is a complementary axis

Cumulative reward measures what a model plays; coupling measures whether its stated reasoning predicts that play. Table 5 reports coupling for the five A+β variants and three frontier references, and Figure 4 plots fc in (ID, HO) space in the same layout as the reward scatter.

Two observations. First, the narrower A+β mixes trade coupling for reward: CORE through ALL drop ID fc to 0.56–0.61 (base: 0.77), and CORE collapses to zero held-out coupling outright. The same two interventions that resolve the reward trade-off also recover coupling — RW and MERGE reach ID fc 0.68 and 0.69 and HO fc 0.46 and 0.50, the best of any trained variant on both final-round columns. Second, coupling and reward order the frontier models differently: Gemma 3–27B has the highest all-round ID coupling (0.80) yet among the worst held-out reward (+0.28), while Haiku leads both. A model whose reasoning faithfully predicts its actions can still play the wrong actions; the two metrics are complementary rather than redundant.

3.6 A falsified precursor: coupling-filtered DPO on frontier actions

We initially hypothesised that filtering DPO pairs by reasoning-action coupling — keeping only trajectories whose actions follow their stated expected values — would teach a small model to bind reasoning to action, with frontier-played actions as labels. The data falsified both halves. With the filter on, ID fc *fell* from 0.77 to 0.66; with it off, to 0.55; ID reward moved flat to down (14.04 to 13.77 and 13.51); and neither variant reached positive held-out reward. We retain this result because it isolates what matters in A+β: the gains of §3.2 appear only once the action label itself is replaced by the solver’s best response, not when frontier actions are filtered more aggressively.

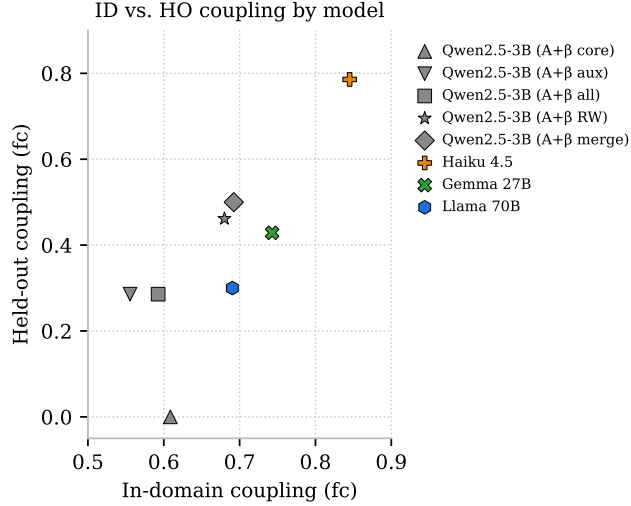


Figure 4: In-domain vs. held-out final-round coupling fc by model, in the same layout and colour scheme as Figure 2 (Qwen 2.5–3B $A+\beta$ variants grey; frontier references coloured). $A+\beta$ -RW (star) and $A+\beta$ -MERGE (diamond) sit highest among the 3B variants on both axes; Haiku 4.5 leads overall, while $A+\beta$ -CORE (triangle) collapses to zero held-out coupling.

Table 5: Reasoning-action coupling on the 12-env eval. fc = final-round coupling (fraction of parsable rounds whose chosen action matches the argmax of the stated [EV]); ac = all-round coupling. Reported for the trained Qwen 2.5–3B $A+\beta$ variants and the frontier references. Best 3B-variant cell per metric in **bold**.

Model	$fc(ID)$	$ac(ID)$	$fc(HO)$
Qwen 2.5–3B + $A+\beta$ -CORE	0.61	0.74	0.00
Qwen 2.5–3B + $A+\beta$ -AUX	0.56	0.72	0.29
Qwen 2.5–3B + $A+\beta$ -ALL	0.59	0.65	0.29
Qwen 2.5–3B + $A+\beta$ -RW	0.68	0.70	0.46
Qwen 2.5–3B + $A+\beta$ -MERGE	0.69	0.73	0.50
Bedrock Haiku 4.5	0.85	0.57	0.79
Gemma 3–27B	0.74	0.80	0.43
Llama 3.3–70B	0.69	0.52	0.30

3.7 Scope and limitations

Three boundaries on these results are worth stating plainly. First, each cell aggregates $n = 12$ episodes per environment against deterministic or low-variance opponents; we report no confidence intervals, and differences below roughly 0.2 CR (such as RW vs. MERGE) should be read as ties. Second, the in-domain coupling gap to Haiku does not close (0.69 vs. 0.85), and on games where Haiku exploits stationary opponents well (pd-classic against always-cooperate) the trained model gives up 1–2 reward points; $A+\beta$ buys held-out and strategic-opponent competence, not uniform dominance. Third, every trained variant shares one base model (Qwen 2.5–3B), one solver family (§2.2), and one frontier paraphraser (Haiku 4.5); whether the recipe transfers across base scales or to games without tractable analytical best responses is untested here.

4 Related Work

LLMs in non-cooperative games. A growing benchmark literature establishes that LLMs fall short of game-theoretic solution concepts in non-cooperative settings beh [2025], gtb [2024], gam [2024b,a], and non [2024] surveys the observed failure modes. These works measure the gap; they do not ask which training-time supervision closes it. Our frontier-reference rows (§3.1) reproduce the

332 gap at per-opponent granularity — isolating, for example, the shared frontier failure against tit-for-tat
333 — and the rest of the paper is about closing it.

334 **Preference optimization.** $A+\beta$ trains with standard sigmoid-loss DPO Rafailov et al. [2023] on
335 top of TRL von Werra et al. [2020], adding environment-token loss masking as a trainer subclass.
336 Alternative preference losses such as SimPO Meng et al. [2024] and IPO Azar et al. [2024] (surveyed
337 in Liu [2024]) are orthogonal to our contribution: we change where the preferred completion *comes*
338 *from*, not how the preference is optimised. Nothing in the recipe depends on the sigmoid loss.

339 **Process supervision and credit assignment.** Process reward models score intermediate reasoning
340 steps Lightman et al. [2024]; our recipe is closer to step-level preference construction, with the
341 difference that the “correct step” is an analytical solver’s best response rather than a learned scorer’s
342 estimate. GRPO Shao et al. [2024] assigns credit in sequential settings by group-normalising K
343 on-policy rollouts. As discussed in §2.4, our horizon-aware counterfactual replaces that stochastic
344 estimate with one exact dynamic-programming pass, at the price of requiring a reconstructible
345 opponent — a trade that favours us at training time and GRPO at deployment.

346 **Model merging.** Averaging our specialist adapters (§2.5) is an instance of task arithmetic Ilharco
347 et al. [2023]. TIES-merging Yadav et al. [2023] and Fisher-weighted merging Matena and Raffel
348 [2022] resolve interference between task vectors; our result is that for two DPO specialists sharing a
349 base model and LoRA configuration, no interference resolution is needed — plain averaging matches
350 a reweighted retrain (§3.4). We are not aware of a prior demonstration of merge-retrain equivalence
351 for strategic-game preference-trained specialists.

352 **Multi-agent LLM systems.** Work on LLM agents interacting with other agents concentrates
353 on cooperative settings owl [2025], mas [2024b,a]; adversarial and mixed-motive interaction is
354 comparatively underexplored adv [2024], Anthropic [2024]. We treat the two-player non-cooperative
355 case as the tractable entry point where analytical solvers exist, exact counterfactuals are computable,
356 and the supervision-provenance question can be answered cleanly before scaling to richer multi-agent
357 settings.

358 5 Discussion and Conclusion

359 **What the results establish.** Frontier LLMs and a 3B base model fail against strategic opponents in
360 the same way and to the same degree, which removes frontier action distillation as a viable recipe for
361 strategic play. Replacing the action label with an analytical solver’s ex-post best response — while
362 keeping the frontier as a reasoning paraphraser and filtering flips by exact counterfactual rollout —
363 recovers 93% of the frontier’s held-out reward from the 3B base and lifts tit-for-tat reward from 1.50
364 to 14.06, a level no frontier reference reaches (§3.2–3.3). The specialist-forgetting trade-off between
365 data mixes dissolves under either of two interventions, and the fact that parameter averaging matches
366 a reweighted retrain (§3.4) means the combined capability costs a tensor sum rather than a training
367 run.

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368 **What the results do not establish.** The in-domain coupling gap to Haiku does not close, and $A+\beta$
369 gives up 1–2 reward points on games where the frontier profitably exploits stationary opponents
370 (§3.7). Negotiation training transferred less cleanly than auction training. The recipe requires game
371 families with tractable analytical best responses and training opponents reconstructible from their
372 names; sequential perfect-information games such as tic-tac-toe (where minimax is the natural solver),
373 stochastic opponents, and open-ended dialogue games all sit outside the demonstrated envelope. The
374 merge result covers two adapters at $\alpha = 0.5$ on one base model; N -specialist, asymmetric, and
375 cross-scale merging are untested, as is the recipe’s behaviour at base scales other than 3B.

376 **Conclusion.** When the goal is strategic play, the binding constraint in our experiments is not model
377 scale but the provenance of the action label: a solver that plays the game correctly is a better action
378 teacher than a frontier model that merely describes correct play. We offer $A+\beta$ as a recipe built
379 on that division of labour — solver actions, frontier reasoning, counterfactual credit — together
380 with a twelve-environment evaluation suite that exposes the frontier’s strategic-opponent failures at
381 per-opponent granularity, as concrete starting points for training small multi-agent-capable LLMs.

Appendix

A. Hypothesis B retrospective: why coupling-filter DPO didn’t work

Before $A+\beta$, we ran a full DPO pipeline (which we call Hypothesis B internally) that pinned chosen-side actions to the frontier model’s recorded play and filtered pairs by reasoning-action coupling. The recipe was:

1. Generate trajectories with frontier (Haiku 4.5) as player 0, heuristic opponents.
2. For each trajectory, parse the [EV] slot in chain-of-thought. Tag a trajectory “coupled” if the played action equals the argmax of the parsed EVs.
3. Build DPO pairs by S1 (outcome / cumulative reward), S2 (NE-aligned action), S3 (cross-strategy consensus), S4 (L2 vs L0 contrastive). Filter chosen-side to coupled-only trajectories.

The headline coupling claim — that filtering teaches small models to bind reasoning to action — did not survive the data. Filter-on Qwen 2.5–3B post-DPO final-round ID coupling went from 0.77 (base) to 0.66 (down). Filter-off went $0.77 \rightarrow 0.55$ (further down). Two cumulative reward observations were also off-prediction: filter-off (-0.23 HO CR) beat filter-on (-0.95), and both lost ID CR vs base. We had predicted the opposite ordering.

Diagnosis: training-distribution narrowness. Five of the eight ID training games (PD $\times$ 3, stag-hunt, BoS) sit on a single “cooperate-or-defect-vs-reciprocity” axis. Three of those five are the same matrix-game stage with different payoff parameters. The opponent pool is similarly concentrated: TFT, AlwaysCoop, AlwaysDefect, GrimTrigger, Pavlov, TitForTwoTats, GenerousTFT, EpsilonGreedy — seven of nine are reciprocity-style. Under such a narrow training mix, DPO overwrites whatever decision habits the base RLHF gave Qwen 2.5-3B with stricter reciprocity patterns. The chosen-side coupling filter narrows the support further (388 pairs vs 407 unfiltered), giving filter-on *less* variety to learn from — the opposite of what the filter was supposed to do.

This diagnosis bridged us to $A+\beta$. Filter-off’s modest HO CR gain ($+1.22$ CR vs base) suggested wider chosen-side distributions generalise better. $A+\beta$ replaces the filter with a different intervention: rather than narrowing chosen-side trajectories to “frontier action that happens to match its own argmax-EV,” it *replaces the action label entirely* with the analytical solver’s best response. The data is then unrelated to the filter mechanism; the question becomes whether better action labels transfer better than filtered frontier-pinned action labels. The answer (§3) is yes.

Why we report this in the paper. Two reasons: (i) the falsified coupling thesis was load-bearing in our initial framing, and reporting only the working method without the prior failure hides where the actual contribution sits; (ii) the diagnostic claim about narrow training distributions causing forgetting is itself a reusable observation — it returned in the AUX-vs-ALL trade-off (§3.4), where it was again resolved by widening the chosen-side distribution rather than narrowing it.

B. Frontier-coupling measurements

Frontier final-round coupling rates by environment, computed by the prose-form-aware parser (Appendix 5). n/a indicates the env had no parsable EV slots in any final round (typical for tic-tac-toe where the prompt does not elicit slot form, and for continuous-action envs).

Two patterns visible. (i) Coupling drops sharply on held-out games for all but Haiku and Gemma. (ii) Llama 3.3-70B has the largest *aggregate* parameter count but lower PD-game coupling than Gemma 3–27B and Llama 3.1–8B — scale alone does not explain coupling.

C. Leakage audit

The $A+\beta$ frontier paraphrase prompt (§2.3) is engineered to write reasoning that does not reference the current-round opponent action as observed. We audit two prompt versions: v1 (initial design, told frontier the opponent’s move directly), v2 (no-leakage rewrite, posterior-EV framing).

Table 6: Per-(model, env) final-round coupling rate, parsable rounds only. $n = 12$ episodes per cell. Game abbreviations: pd-c = pd-classic, pd-t = pd-tight, pd-h = pd-high-temptation, mp = matching-pennies, ttt = tic-tac-toe, auct = auction, dd = divide-dollar, p-b = p -beauty, ipd = ipd-stage.

Model	pd-c	pd-t	pd-h	stag	nego	bos	mp	ttt	auct	dd	p-b	ipd
Haiku 4.5	0.87	0.90	0.91	0.89	0.00	1.00	1.00	n/a	0.00	n/a	0.00	0.92
Llama 3.1–8B (B)	0.90	0.43	1.00	0.43	0.00	1.00	0.71	0.00	n/a	0.00	n/a	0.67
Gemma 3–27B	0.38	0.75	0.83	0.91	0.00	1.00	0.91	0.00	n/a	0.00	n/a	1.00
Llama 3.3–70B	0.60	0.33	0.80	0.50	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.75
Qwen 2.5–3B base	1.00	0.80	0.83	0.75	n/a	0.78	0.75	0.00	n/a	0.00	0.00	1.00
Qwen 2.5–3B + A+ β -on	0.83	0.33	0.50	1.00	0.00	1.00	0.60	n/a	0.00	0.00	0.00	0.67
Qwen 2.5–3B + A+ β -off	0.25	0.33	0.33	0.00	0.00	0.88	0.88	0.00	n/a	n/a	0.00	0.83

v1 leakage rate. On a sample of 4 round-1 oracle traces across 7 envs, all 4 round-1 traces wrote phrases like “The opponent played C in round 1” or “opponent played Stag this round”. v1 was unsafe to train on: the small model would learn to hallucinate observed-this-round actions at deployment.

v2 leakage rate. Across all 688 oracle rounds in the final batches (112 round-1 + 576 later rounds), the leakage regex `r"opponent (played|chose|cooperat|defect)\w*\s+(this round|in round \d+)"` matches 7 cases. Manual inspection confirms all 7 are false positives discussing prior rounds (e.g. “Round 1 history: I played C, opponent played C”). The round-1-specific regex `r"opp(onent)? play(ed|s)|opp(onent)? cooperat|opp(onent)? defect"` matches 6 round-1 traces; 5 are false positives discussing hypothetical opponent types (“opponent could be a cooperator”), 1 is a model echoing the system prompt’s own prohibition.

Empirical conclusion. v2 leakage rate is effectively 0. The +1.91 HO CR gain from A+ β -RW is not explained by hindsight hallucination at training time.

D. Coupling parser

We parse chain-of-thought for an [EV] slot and extract per-action expected values via three regex patterns. All patterns require an action label, an arrow/colon/equals separator, an arithmetic expression with at least one operator, and a final equals-result number.

- **Pattern A** (slot form): bare action label + colon or arrow. Examples: “C: 0.5*3 + 0.5*0 = 1.5”, “Stag → 0.5*4 = 2.0”, “**Action C:** 0.5*3 = 1.5”.
- **Pattern B** (EV-wrapper form): “EV(C) = 0.30(3.0) + 0.40(0) = 1.2”. Common in Anthropic outputs.
- **Pattern C** (prose form, post-DPO): “Expected payoff for ‘C’ is 0.6*3 + 0.4*0 = 1.8”, “Total EV for ‘D’ = 1.5 + 0.7 = 2.2”. Trained Qwen models tend to produce this form.

After regex match, an action label is dropped if it appears in a list of English-prose words mistakenly captured (round, rounds, value, playing, etc.).

A round is *coupled* iff the action with the maximum stated EV equals the chosen action. A round is *parsable* if at least one EV expression matches; rounds without parsable EVs are excluded from the denominator. Final-round coupling restricts to the last round of each episode.

Released as `rl_noncoop/scoring/coupling.py` alongside 17 unit tests (matrix-game stages, prose-form Qwen outputs, edge cases). Adding Pattern C lifted parsable-round counts on Qwen 2.5–3B PD outputs by $\sim 3\times$ on average, exposing real coupling rates that the slot-only parser had missed.

E. Training hyperparameters

All trained A+ β variants use Qwen 2.5–3B-Instruct, LoRA with $r = 16$, $\alpha = 32$, dropout 0.05, target modules auto, max sequence length 12288. Training: 10 epochs, per-device batch 1, gradient accumulation 8 (effective batch 8), learning rate 5×10^{-5} cosine schedule with 5% warmup, gradient

464 checkpointing on, no flash attention, sigmoid DPO loss with $\beta = 0.1$, environment-token loss
 465 masking on, save every 10 steps, `load_best_model_at_end=true`. Single L40S GPU, 13–14 GB
 466 peak memory. The Hypothesis B variants share the same recipe.

Variant	Pairs	Best ckpt step	Best eval loss
B filter-on	388	60	0.3734
B filter-off	407	110	0.2721
A+ β -CORE	503	50	0.2973
A+ β -AUX	613	80	0.3128
A+ β -ALL	1338	40	0.2278
A+ β -RW	1749	70	0.2454
A+ β -MERGE	—	—	—

Table 7: Training hyperparameters consistent across all DPO variants except for pair count and best checkpoint. A+ β -MERGE is not a trained variant; it is the LoRA-average of AUX and ALL adapters.

467 E. Two example A+ β pairs

468 We include two pedagogical examples from `runs/oracle_batch1_*` to illustrate the kind of training
 469 signal A+ β produces.

470 **Example 1 — pd-classic, episode 0, round 10 (final round).** Iterated PD with $T = 5/R =$
 471 $3/P = 1/S = 0$, opponent = TFT. Both sides cooperated for 9 rounds. Round 10 is the final
 472 round; no shadow of the future. Blind player 0 played C (cumulative reward 30.0) despite computing
 473 $EV(D) = 4.95 > EV(C) = 3.0$ in its own [EV] slot.

474 **Blind reasoning [Decision] excerpt:**

475 [DECISION] Play C. While defection yields 5 vs 3 this round, the opponent’s 9-round perfect
 476 cooperation record and the statement that “the game continues” suggest either: (1) there are
 477 more rounds coming, or (2) reputation matters. Defecting would break a mutually beneficial
 478 pattern.

479 **A+ β oracle reasoning [EV] and [Decision] excerpts (counterfactual reward 32.0, +2):**

480 [EV] This is the final round ($T=10$), so there is no future to consider. C: 3.000 (mutual
 481 cooperation payoff). D: 5.000 (defection payoff when opponent cooperates).
 482 [DECISION] Play D. In the final round of a finite repeated game, the standard solution is to
 483 defect regardless of history; the opponent cannot retaliate.

484 The blind trace’s reasoning was *coupled-internal* (it computed EVs and chose argmax) until the
 485 [Decision] step — where it overrode the EV with a horizon argument that does not apply to the last
 486 round. Hypothesis B’s coupling filter would not have caught this: the EV computation matched the
 487 chosen action up to the very last sentence. A+ β ’s solver says final-round-stage-best-response is D,
 488 the counterfactual confirms +2 CR over what blind achieved, and the pair is emitted with the oracle
 489 excerpt as chosen and the blind excerpt as rejected.

490 **Example 2 — stag-hunt, episode 2, anti-coordination.** 10-round stag-hunt with payoffs
 491 (Stag,Stag)= 4, (S,H)= 0, (H,S)= 3, (H,H)= 1, opponent = TFT. Blind player 0 was *anti-*
 492 *coordinating* with TFT (Table 8). Blind cumulative reward was 19.0; A+ β ’s oracle would have
 493 achieved 24.0 across the 5 helps-flagged rounds.

494 Blind’s [Prior] for round 1 placed 0.4 probability on an exploitative opponent type, leading to a
 495 Hare-as-risk-dominant decision. The actual opponent (TFT) was zero-probability exploitative. This
 496 is the failure mode Hypothesis B’s coupling filter literally cannot reach: blind’s EV-action coupling
 497 for that round was 1.0 (it chose argmax of its own stated EVs); the mistake was one level up, in the
 498 prior. A+ β factors out prior-formation by giving the small model a solver-pinned answer: against
 499 TFT-played-Stag, best response is Stag. The small model learns from supervision; it does not have to
 500 derive the prior from scratch.

round	TFT	blind	oracle
1	Stag	Hare	Stag (HELPS)
2	Hare	Stag	Hare (HELPS)
3	Stag	Hare	Stag (HELPS)
6	Hare	Stag	Hare (HELPS)
7	Stag	Hare	Stag (HELPS)

Table 8: Stag-hunt anti-coordination. Blind p0 played the opposite of TFT’s previous action each round. Other rounds (4, 5, 8, 9, 10) agreed.

501 G. Per-(model, env) coupling, exploitability, OPA

502 The full per-cell breakdown is large (96 rows: 8 reference models + DPO variants $\times$ 12 envs).
503 We reproduce the Haiku 4.5 row here to anchor the reader; the complete table is available in the
504 companion repository.

Haiku 4.5 / env	ac	fc	exploitability
pd-classic	0.44	0.91	0.73
pd-tight	0.47	0.90	0.72
pd-high-temptation	0.54	0.91	0.58
stag-hunt	0.85	0.89	0.27
negotiation	0.00	0.00	n/a
bos	1.00	1.00	0.00
matching-pennies	1.00	1.00	1.00
tic-tac-toe	0.00	n/a	n/a
auction	0.00	0.00	n/a
divide-dollar	n/a	n/a	n/a
p-beauty	0.00	0.00	n/a
ipd-stage	0.92	0.92	0.00

Table 9: Haiku 4.5 per-env companion metrics. ac = all-round coupling, fc = final-round coupling. Exploitability via maxmin LP for matrix games; n/a for continuous-action and one-shot games where the metric is undefined or trivially zero.

505 H. Opponent pools by environment family

506 **Repeated 2×2 matrix games (PD $\times$ 3, stag-hunt).** 9 deterministic heuristics with weighted
507 sampling: TFT (weight 2.0), GrimTrigger (2.0), TitForTwoTats (1.5), GenerousTFT (1.5), Pavlov
508 (1.5), AlwaysCooperate (1.0), AlwaysDefect (1.0), RandomPolicy (1.0), EpsilonGreedy (1.0). Per-env
509 opponent sub-selection in our YAMLS.

510 **First-price auction.** Four bidder heuristics with round-robin warm-up sampling:
511 `truthful_bidder` ($b=v$), `shaded_bidder_low` ($b=0.3v$), `shaded_bidder_high` ($b=0.7v$),
512 `random_uniform_bidder` ($b \sim \text{Unif}[0, v]$).

513 **Multi-issue bargaining.** Four bargainer heuristics with round-robin warm-up:
514 `greedy_bargainer` (claims capacity on every issue), `equal_split_bargainer` (claims
515 half capacity each), `nash_bargainer` (claims all of own highest-weight issue, half elsewhere),
516 `random_bargainer` (uniform random proposal in $[0, c]^3$).

517 **Round-robin warm-up.** For all single-shot game families, we replaced the original purely-
518 weighted-random opponent sampler with a round-robin warm-up that ensures every opponent receives
519 at least one episode in the first $|\text{pool}|$ draws. This was a fix for a sampler bug discovered in our
520 second batch where TFT drew zero episodes despite a non-zero weight; the round-robin guarantees
521 minimum coverage at small batch sizes.

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